Preparing for the Finals

The slides here only outlines the important algorithms, but we have covered in class more than what is reflected here.

Mostly problem-solving type of questions

TOPICS COVERED IN THE EXAM

1. State-Space Search (12 marks)
2. CSP (10 marks)
3. Games (10 marks)
4. Fuzzy Logic (10 marks)
5. Neural Networks (10 marks)
6. Propositional Logic (8 marks)

Total = (60 marks)

Come up with a strategy for budgeting your time to answer all questions in the exam.

Final Exam Preparation (2023)

Please see the section, titled “Final Exam Preparation (2023)” in our stream website to find the following useful materials:

1. Sample past exam
2. E-book 159302 – to be used as reference in the exam

On-line Supervised Exam (OSE)

- The exam will be conducted **online and will be supervised by Massey's Assessment Services Team**.
- **NO part of the exam may be retained or shared for any other purpose**. Doing so would constitute cheating and appropriate penalties will apply.
- The exam is a combination of **multiple-choice and fill-in the blanks** type.
- **You are allowed to utilise an eBook (159302) during the exam.** This e-book is a compilation of our lecture slides, tutorials and tutorial solutions. **This e-book will be made accessible within the exam itself**; a clickable link is provided in the online exam to open the e-book. No other references are allowed by Massey's Assessment Services Team apart from this e-book. You will find a copy of the e-book in our stream website to get familiarised.

On-line Supervised Exam (OSE)

- You will need a **scientific calculator (no restrictions)** to answer some questions in the exam.
- You are allowed to use **six (6) sheets of blank A4 paper** (as scratch papers) to perform your calculations. These scratch papers will not be submitted after the exam. Massey's OSE impose some restrictions.
- You might want to use a **pencil and a "nice" big eraser** in the exam so that you will not run out of space in your scratch papers.
- The exam is **time-constrained**. You will be given **3 hours** to complete and submit the exam.

On-line Supervised Exam (OSE)

- Since there are students enrolled in the course who are overseas, I am going to allow the class to choose a time to take the exam from 9:30am (NZ time) on the 6th of November, up to 9:30am (NZ time) on the 7th of November. **Massey's Assessment Services advises taking the exam during their support hours.** This is what they say:

During **support hours of 8am – 6pm NZST** contact Assessment Services:
Phone: **0800 627 739**; or from overseas **+64 6 350 5701**

- or –

Email: exams@massey.ac.nz. Provide **your Student ID and Course number** and outline the problem

- Massey's Assessment Services are the ones in charge of conducting the Online Supervised Exam.

Online Supervised Exam

Let’s have a look at how the exam looks like.

QUIZ

159302 Semester 2 Final Exam

Date: **6th November 2023 starting at 9.30am**

Time allowed is **three (3)** Hours

NO part of this examination may be captured unless you are specifically instructed to download and save a file.

NO part of the examination may be retained or shared. Doing so would constitute cheating and appropriate penalties will apply.

This is an Online Supervised Examination (OSE).

- You are permitted to have 6 sheets of blank A4 paper and a pen to make notes and calculations. Show all twelve sides of the papers to the camera during the room scan part of the PSI login. These pages do not need to be uploaded at the end of your exam.
- Calculators are permitted with no restrictions. Please show your calculator to the camera during the room scan part of the PSI login.

Opens 6 November 2023, 9:30 AM NZDT **Closes 7 November 2023, 9:30 AM NZDT**

Sample Questions

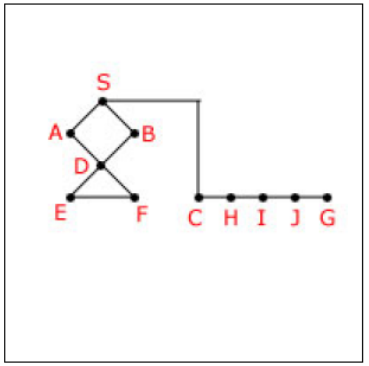
Click here to access **159302 e-book** as your reference

You will be able to open a pdf file as a reference during the exam.

Search

We are trying to find a path from **S** to **G** in the following graph:

```
Connections (all undirected):
 S: A B C            F: D E
 A: S D              G: J
 B: S D              H: C I
 C: S H              I: H J
 D: A B E F          J: I G
 E: D F
```



The heuristic distance estimates to the goal are:
S = 10, A = 5, B = 8, C = 6, D = 7, E = 6,
F = 5, H = 3, I = 2, J = 1, G = 0

Examine states in alphabetical order when no other order is specified. In particular, assume that the list of descendant states of a parent state is placed on Q so that they are examined in alphabetical order. So, this determines the order that both depth first and breadth first explore the space. Also, if you have ties in heuristic searching, use alphabetical order to break ties.

1. Apply A* (using strict Expanded List) to solve the problem. [4 marks]

Search from node **S** to node **G.**

Write the **sequence of states expanded** by the algorithm, following the format given below.

- For each node expanded, indicate the **f-cost** and the **state name** for each step in the sequence. Use hyphen as delimiter. Spaces are not allowed in your answer. Do not put any spaces in between the nodes and hyphens.
- Sample Answer: 0S-8C-7B-25G

Some questions require a specific format for writing your answers.

Answer:

Click here to access **159302 e-book** as your reference

CSPs

Simple Map Colouring Problem

Let's look at a simple map coloring problem and explore the behavior of constraint propagation, backtracking and forward checking in this context. Here is the relevant problem specification:

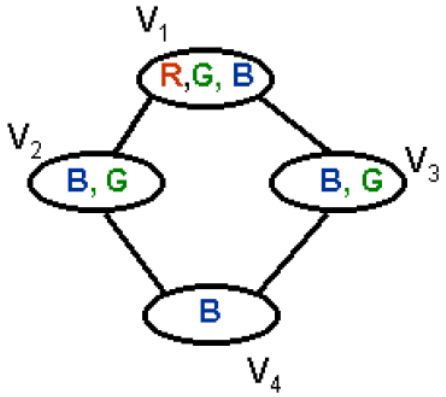


Figure: Constraint Graph 1.

1. Refer to Constraint Network 1 (see figure). Apply the **Backtracking with Forward Checking algorithm**. Use a fixed variable ordering in the search (V1, V2, V3, V4) and consider the values in the order shown in the diagram. Stop when you reach the first solution, or the end of the search tree if no solution exists.

[5 marks]

Write the sequence of assignment of values to variables using the following format:

- For each step in the sequence, indicate the variable name and the value assigned for the variable. Use hyphen as delimiter.
- Spaces are not allowed in your answer. Do not put any spaces in between the variable names, assigned values and hyphens.

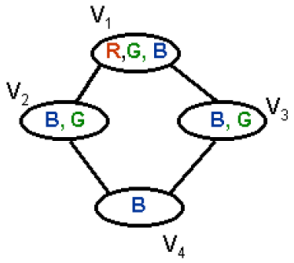
Sample answer: V1R-V2B-V2G-V3B-V4B

Some questions require a specific format for writing your answers.

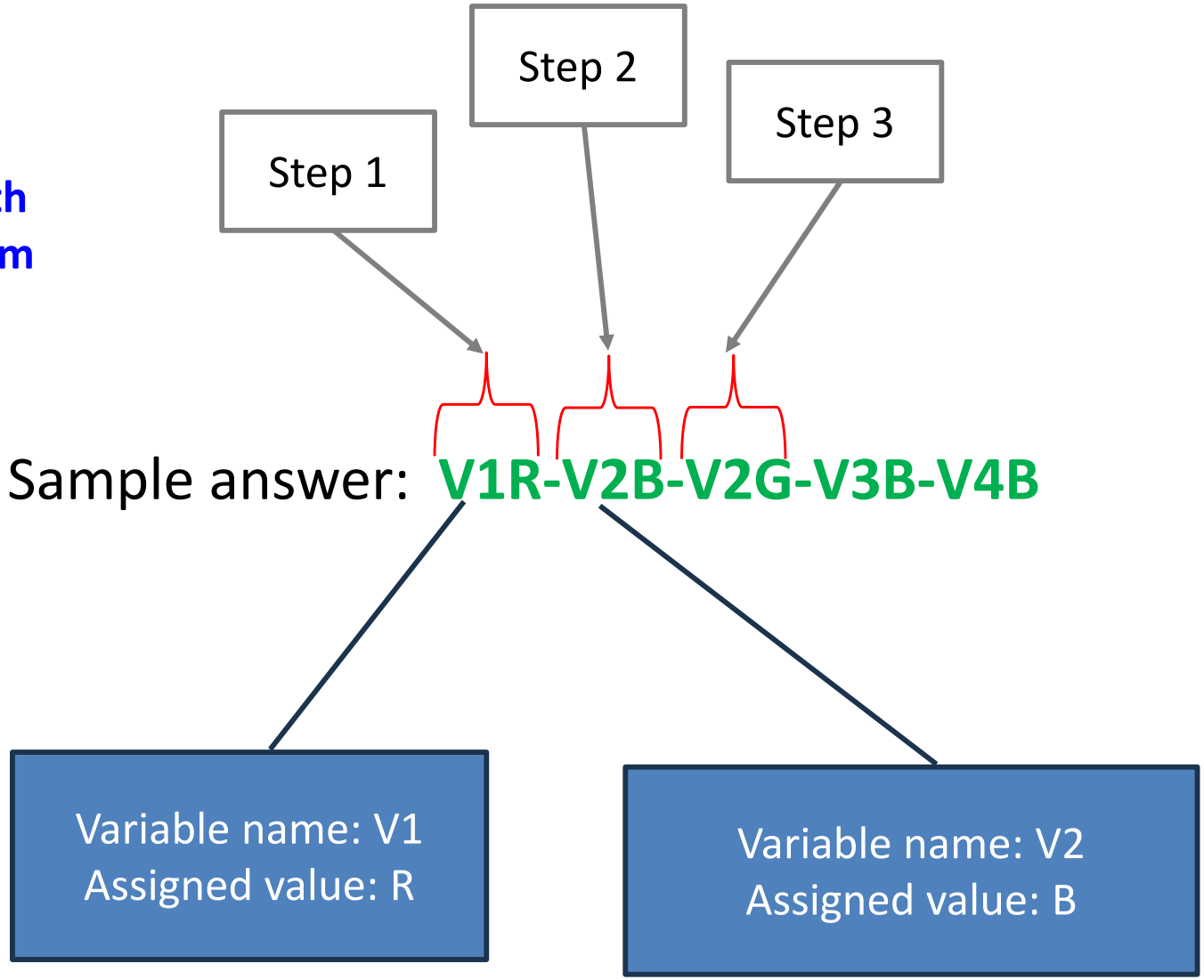
Answer:

Simple Map Colouring Problem

Let's look at a simple map coloring problem and explore the behavior of constraint propagation, backtracking and forward checking in this context. Here is the relevant problem specification:



Apply the **Backtracking with Forward Checking algorithm**



Click here to access **159302 e-book** as your reference

Alpha-Beta Pruning (left to right)

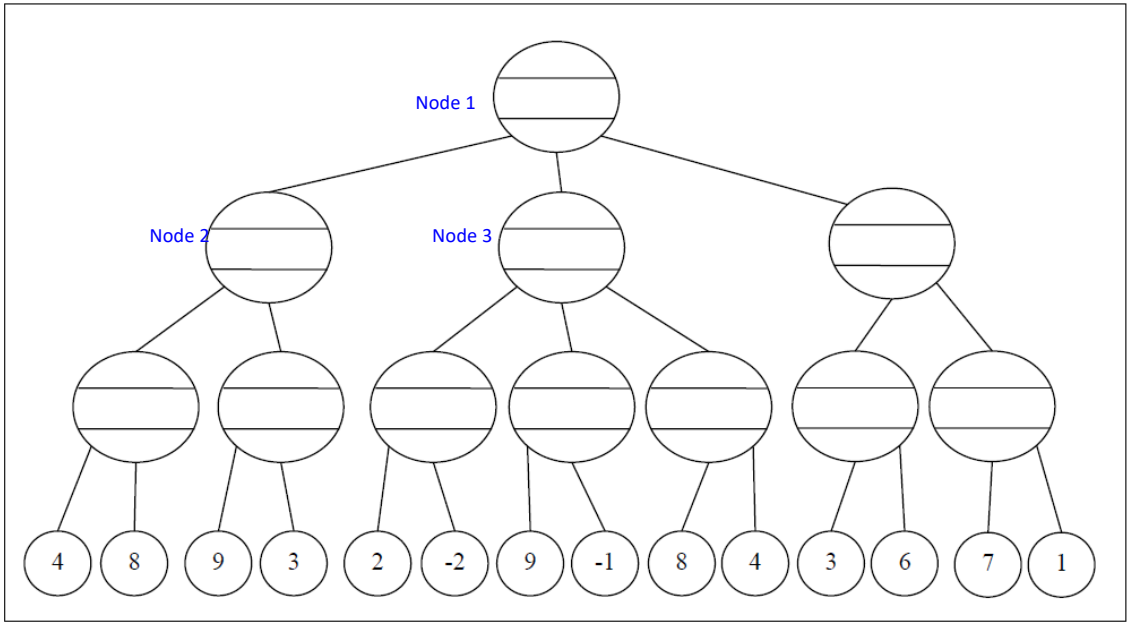
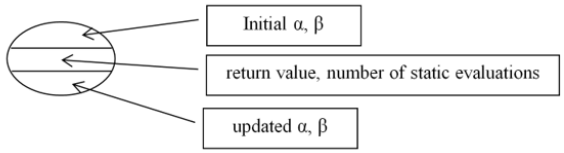


Figure: Game Tree



Node Interior Structure

Simulate **Alpha-Beta Pruning** on the given game tree. As in the lectures, use the given node structure to keep track of the details of the computations. Answer the following questions based on the results of your simulation.

Question: Refer to the Game Tree on the figure. Based on the results of your simulation, what are the initial alpha and beta values assigned to Node 3? [1 mark]

a. (-infinity, +infinity)
b. (5, 8)
c. (6, 9)
d. (-2, +infinity)
e. None of the options given are correct